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Medical image processing apparatus, medical image processing system, medical image processing method, and program

Abstract

An apparatus, a system, a method, and a program for medical image processing are provided. The apparatus includes one or more processors configured to acquire an endoscopic image generated through imaging of a living body, perform classification of lesion regions contained in the endoscopic image into two or more classes, identify a classification contributing region in the endoscopic image, the classification contributing region contributing, with a degree of contribution, to the classification of one of the lesion regions contained in the endoscopic image, generate a region image displaying a region in the endoscopic image, the region image displaying the classification contributing region with a density or a heat map according to the degree of contribution, and display the region image along with the endoscopic image on a monitor, where the region image is displayed at a position different from a position at which the endoscopic image is displayed.

Inventors:	Kamon; Shumpei (Kanagawa, JP)
Applicant:	FUJIFILM Corporation (Tokyo, JP)
Family ID:	69142574
Assignee:	FUJIFILM Corporation (Tokyo, JP)
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Primary Examiner: Nguyen; Khai M

Attorney, Agent or Firm: JCIPRNET

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of and claims the priority benefit of a prior application Ser. No. 17/118,535 filed on Dec. 10, 2020, now allowed. The prior application Ser. No. 17/118,535 is a Continuation of PCT International Application No. PCT/JP2019/023696 filed on Jun. 14, 2019 claiming priority under 35 U.S.C § 119(a) to Japanese Patent Application No. 2018-130138 filed on Jul. 9, 2018. Each of the above applications is hereby expressly incorporated by reference, in its entirety, into the present application.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to a medical image processing apparatus, a medical image processing system, a medical image processing method, and a program, and more particularly to automatic classification of a lesion in a medical image.

2. Description of the Related Art

(2) In the medical field, examinations using a modality such as an endoscope system are carried out. In recent years, there has been known a technique of analyzing medical images such as endoscopic images which are time-series images captured using an endoscope scope, automatically classifying a lesion included in the endoscopic images, and providing a classification result. Note that classification of images and discrimination of images are treated as the same concept herein.

(3) JP5528255B describes an endoscopic image processing system that assists detection of a characteristic lesion site in an image captured with an endoscope. The endoscopic image processing system described in JP5528255B displays a dotted line so as to surround a location estimated to be a lesion site in an image displayed on a monitor when a lesion estimation function is executed.

(4) JP2017-70609A describes an image processing system that performs texture analysis on the density of the entirety of a captured image of a blood vessel, and uses the result of the texture analysis to perform classification corresponding to pathological diagnoses of a non-tumor, an adenoma, and the like. JP2017-70609A describes an example of displaying a probability of the classification being correct.

(5) JP4615963B describes an endoscope apparatus that removes an image inappropriate for diagnosis from among images of the inside of a body cavity captured using the endoscope apparatus.

(6) JP5576711B describes an image processing apparatus that processes time-series images of an in-vivo lumen captured in time series using a medical observation apparatus. The image processing

apparatus described in JP5576711B determines, as a specific region, a region of normal mucosa in an image constituting the time-series images. The image processing apparatus described in JP5576711B also calculates a degree of reliability of the specific region.

SUMMARY OF THE INVENTION

(7) When the type of a lesion found in an examination of the inside of a body cavity performed using an endoscope apparatus is automatically classified using artificial intelligence (AI), the AI that performs automatic classification receives images acquired during the examination and outputs a classification result.

(8) However, the images acquired during the examination include images that are not suitable for classification using the AI, such as an image in which a part of a lesion is not depicted because the lesion is hidden by an object although the lesion is present and an image in which a lesion is depicted but is blurred. In such a case, the AI may output an inappropriate classification result.

(9) With the inventions described in JP5528255B, JP2017-70609A, JP4615963B, and JP5576711B, when an image that is not suitable for automatic classification described above is classified, an inappropriate classification result may be output. On the other hand, it is difficult for an observer to determine whether the classification result is appropriate.

(10) The present invention has been made in view of such circumstances, and an object of the present invention is to provide a medical image processing apparatus, a medical image processing system, a medical image processing method, and a program that allow an observer to grasp whether classification result is appropriate in automatic classification of a medical image.

(11) In order to accomplish the object described above, the following aspects of the invention are provided.

(12) A medical image processing apparatus according to a first aspect includes: an image acquisition unit that acquires a captured image generated through imaging of a living body; a classification unit that classifies the captured image into two or more classes; an image generation unit that generates a region image depicting a location of a region that contributes to classification performed using the classification unit, in the captured image subjected to the classification performed using the classification unit; and a display signal transmission unit that transmits to a display device, a display signal representing the captured image, a classification result derived using the classification unit, and the region image. The display signal transmission unit transmits, to the display device, a display signal for displaying the region image separately from the captured image.

(13) According to the first aspect, the captured image is classified into two or more classes, the region image depicting the region that contributes to the classification in the captured image is generated, and the region image is displayed separately from the captured image using the display device. This thus allows an observer to grasp a region that contributes to classification and to grasp whether the classification is appropriate.

(14) A medical image captured using medical equipment may be used as the captured image. The medical image refers to a captured image of a living body generated using a modality such as an endoscope apparatus, a computed tomography (CT) apparatus, a magnetic resonance imaging (MRI) apparatus, or an X-ray imaging apparatus.

(15) Predetermined medical classifications may be used as the classes.

(16) A second aspect may be configured such that in the medical image processing apparatus according to the first aspect, the image generation unit changes a depicting manner of the region image in accordance with the classification result of the captured image.

(17) According to the second aspect, it becomes easier to recognize the distinction between the classifications.

(18) Examples of the depicting manner include a depicting manner using color and a depicting manner in which the location is changed. A depicting manner using both of these depicting manners may also be used.

- (19) A third aspect may be configured such that in the medical image processing apparatus according to the first or second aspect, the classification unit classifies the captured image on the basis of a feature quantity acquired from the captured image, and the image generation unit generates the region image on the basis of the feature quantity.
- (20) According to the third aspect, the region image based on the feature quantity of the captured image can be generated.
- (21) In the third aspect, the captured image may be divided into a plurality of regions, a feature quantity may be calculated for each of the regions, and a region that contributes to classification in the captured image may be identified on the basis of the features of the respective regions.
- (22) A fourth aspect may be configured such that in the medical image processing apparatus according to the first or second aspect, the classification unit employs a deep learning device that has been trained, and the image generation unit generates the region image on the basis of information of an intermediate layer of the deep learning device.
- (23) According to the fourth aspect, the region image based on the information of the intermediate layer of the trained deep learning device can be generated.
- (24) A fifth aspect may be configured such that in the medical image processing apparatus according to any one of the first to fourth aspects, the classification unit calculates, for each of a plurality of regions set in the captured image, membership degrees for the classes, and classifies the captured image on the basis of the membership degrees.
- (25) According to the fifth aspect, classification based on the membership degrees can be performed.
- (26) Examples of the membership degrees include membership probabilities for the classes and scores for the classes.
- (27) A sixth aspect may be configured such that in the medical image processing apparatus according to the fifth aspect, the image generation unit generates the region image on the basis of the membership degrees.
- (28) According to the sixth aspect, the region image based on the membership degrees can be generated.
- (29) A seventh aspect may be configured such that in the medical image processing apparatus according to any one of the first to sixth aspects, the classification unit performs exception determination for the classification based on the region image, and the display signal transmission unit transmits, to the display device, a display signal representing a result of the exception determination performed using the classification unit.
- (30) According to the seventh aspect, output of an inappropriate classification result may be suppressed for a captured image that is difficult to classify.
- (31) The display signal transmission unit may transmit, to the display device, a display signal representing a result of the exception determination instead of the classification result. The display signal transmission unit may transmit, to the display device, a display signal representing the classification result and a result of the exception determination.
- (32) An eighth aspect may be configured such that in the medical image processing apparatus according to the seventh aspect, the classification unit calculates, on the basis of the region image, a degree of reliability of the classification result derived using the classification unit, and the display signal transmission unit transmits, to the display device, a display signal representing the degree of reliability.
- (33) According to the eighth aspect, the degree of reliability of classification may be grasped.
- (34) A ninth aspect may be configured such that the medical image processing apparatus according to the eighth aspect further includes: a storage instruction acquisition unit that acquires an instruction to store the captured image; and a storage unit that stores the captured image. The storage unit associates, in storing the captured image in the storage unit in accordance with the instruction to store the captured image, at least any of the classification result, the result of the

exception determination, or the degree of reliability of the classification result with the captured image.

(35) According to the ninth aspect, the captured image and the information associated with the captured image may be used. In addition, the information associated with the captured image may be checked.

(36) A tenth aspect may be configured such that in the medical image processing apparatus according to any one of the first to ninth aspects, the display signal transmission unit transmits to the display device, a display signal representing text information of the classification result.

(37) According to the tenth aspect, the classification result may be grasped on the basis of the text information.

(38) The text information may be in a language of any kind. An abbreviation may be used as the text information.

(39) A medical image processing system according to an eleventh aspect includes: an image acquisition unit that acquires a captured image generated through imaging of a living body; a classification unit that classifies the captured image into two or more classes; an image generation unit that generates a region image depicting a location of a region that contributes to classification performed using the classification unit, in the captured image subjected to the classification performed using the classification unit; a display device that displays the captured image and the region image, and a display signal transmission unit that transmits, to the display device, a display signal representing the captured image and the region image. The display signal transmission unit transmits, to the display device, a display signal for displaying the region image separately from the captured image.

(40) According to the eleventh aspect, substantially the same advantages as those of the first aspect can be obtained.

(41) The eleventh aspect may be appropriately combined with any of features that are substantially the same as those specified in the second to tenth aspects. In such a case, a constituent element responsible for a process or function specified in the medical image processing apparatus can be grasped as a constituent element responsible for the corresponding process or function in the medical image processing system.

(42) A medical image processing method according to a twelfth aspect includes: an image acquisition step of acquiring a captured image generated through imaging of a living body; a classification step of classifying the captured image into two or more classes; an image generation step of generating a region image depicting a location of a region that contributes to classification performed in the classification step, in the captured image subjected to the classification performed in the classification step; and a display signal transmission step of transmitting, to a display device, a display signal representing the captured image, a classification result derived in the classification step, and the region image. The display signal transmission step transmits, to the display device, a display signal for displaying the region image separately from the captured image.

(43) According to the twelfth aspect, substantially the same advantages as those of the first aspect can be obtained.

(44) The twelfth aspect may be appropriately combined with any of features that are substantially the same as those specified in the second to tenth aspects. In such a case, a constituent element responsible for a process or function specified in the medical image processing apparatus can be grasped as a constituent element responsible for the corresponding process or function in the medical image processing method.

(45) A program according to a thirteenth aspect causes a computer to implement: an image acquisition function that acquires a captured image generated through imaging of a living body; a classification function that classifies the captured image into two or more classes; an image generation function that generates a region image depicting a location of a region that contributes to classification performed using the classification function, in the captured image subjected to the

classification performed using the classification function; and a display signal transmission function that transmits, to a display device, a display signal representing the captured image, a classification result derived using the classification function, and the region image. The display signal transmission function is configured to transmit, to the display device, a display signal for displaying the region image separately from the captured image.

(46) According to the thirteenth aspect, substantially the same advantages as those of the first aspect can be obtained.

(47) The thirteenth aspect may be appropriately combined with any of features that are substantially the same as those specified in the second to tenth aspects. In such a case, a constituent element responsible for a process or function specified in the medical image processing apparatus can be grasped as a constituent element responsible for the corresponding process or function in the program.

(48) According to the present invention, the captured image is classified into two or more classes, the region image depicting the region that contributes to classification in the captured image is generated, and the region image is displayed separately from the captured image using the display device. This thus allows an observer to grasp a region that contributes to classification and to grasp whether the classification is appropriate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is an overall configuration diagram of an endoscope system including a medical image processing apparatus according to embodiments;

(2) FIG. 2 is a block diagram illustrating a hardware configuration of the medical image processing apparatus;

(3) FIG. 3 is a functional block diagram of the medical image processing apparatus according to a first embodiment;

(4) FIG. 4 is an explanatory diagram of a display screen used in the first embodiment;

(5) FIG. 5 is an explanatory diagram of a display screen according to a comparative example;

(6) FIG. 6 is an explanatory diagram of a display screen displayed when the reliability of a classification result is low;

(7) FIG. 7 is an explanatory diagram of a display screen of another example displayed when the reliability of the classification result is low;

(8) FIG. 8 is an explanatory diagram of a display screen displayed when a plurality of lesions are present;

(9) FIG. 9 is an explanatory diagram of a display example of a classification result;

(10) FIG. 10 is an explanatory diagram of an example of displaying membership probabilities as classification results;

(11) FIG. 11 is an explanatory diagram of an example of displaying scores as classification results;

(12) FIG. 12 is a flowchart illustrating a procedure of a medical image processing method;

(13) FIG. 13 is an explanatory diagram of a display screen according to a modification of a region image;

(14) FIG. 14 is an explanatory diagram of a display screen according to a first modification of the classification result;

(15) FIG. 15 is an explanatory diagram of a display screen according to a second modification of the classification result;

(16) FIG. 16 is an explanatory diagram of a region image to which density according to a degree of contribution is applied;

(17) FIG. 17 is an explanatory diagram of a region image to which a heat map according to the

degree of contribution is applied;

(18) FIG. **18** is a block diagram illustrating an example of a configuration of a classification unit that employs a convolutional neural network;

(19) FIG. **19** is a conceptual diagram of the shape transition of a feature quantity in the convolutional neural network;

(20) FIG. **20** is a conceptual diagram of depiction based on information of an intermediate layer of the convolutional neural network;

(21) FIG. **21** is an explanatory diagram of a segmentation technique;

(22) FIG. **22** is an explanatory diagram of a display screen indicating the classification result of “undetectable”;

(23) FIG. **23** is an explanatory diagram of a display screen indicating the classification result of “undeterminable”; and

(24) FIG. **24** is a functional block diagram of a medical image processing apparatus according to a third embodiment.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

(25) Preferred embodiments of the present invention will be described in detail below in accordance with the accompanying drawings. The same constituent elements are denoted by the same reference signs herein, and redundant description will be appropriately omitted.

(26) Overall Configuration of Endoscope System

(27) FIG. **1** is an overall configuration diagram of an endoscope system including a medical image processing apparatus according to embodiments. An endoscope system **9** illustrated in FIG. **1** includes an endoscope **10**, a light source device **11**, a processor device **12**, a display device **13**, a medical image processing apparatus **14**, an input device **15**, and a monitor device **16**.

(28) The endoscope **10** is an electronic endoscope. The endoscope **10** is also a flexible endoscope. The endoscope **10** includes an insertion section **20**, an operation section **21**, and a universal cord **22**. The insertion section **20** is inserted into a subject. The entire insertion section **20** is formed to have an elongated shape with a small diameter.

(29) The insertion section **20** includes a soft part **25**, a bending part **26**, and a tip part **27**. The soft part **25**, the bending part **26**, and the tip part **27** are coupled to each other to constitute the insertion section **20**. The soft part **25** has flexibility sequentially from a proximal end side to a distal end side of the insertion section **20**. The bending part **26** has a structure that is bendable when the operation section **21** is operated. The tip part **27** includes an imaging optical system (not illustrated), an imaging element **28**, and so on.

(30) A CMOS imaging element or a CCD imaging element is used as the imaging element **28**. CMOS is an abbreviation for complementary metal oxide semiconductor. CCD is an abbreviation for charge coupled device.

(31) An observation window (not illustrated) is disposed on a tip surface **27a** of the tip part **27**. The observation window is an opening formed on the tip surface **27a** of the tip part **27**. A cover (not illustrated) is attached to the observation window. The imaging optical system (not illustrated) is disposed behind the observation window. Image light of a site to be observed is incident onto an imaging surface of the imaging element **28** through the observation window, the imaging optical system, and so on. The imaging element **28** images the image light of the site to be observed incident onto the imaging surface of the imaging element **28** and outputs an imaging signal. The term “imaging” used herein includes the meaning of converting light reflected off from a site to be observed into an electric signal.

(32) The operation section **21** is coupled to the proximal end side of the insertion section **20**. The operation section **21** includes various operating members to be operated by a technician. Specifically, the operation section **21** includes two types of bending operation knobs **29**. The bending operation knobs **29** are used to perform an operation of bending the bending part **26**. Note that the technician may also be referred to as a doctor, an operator, an observer, a user, or the like.

(33) The operation section **21** includes an air/water supply button **30** and a suction button **31**. The air/water supply button **30** is used when the technician performs an air/water supply operation. The suction button **31** is used when the technician performs a suction operation.

(34) The operation section **21** includes a still image capturing instruction part **32** and a treatment tool introduction port **33**. The still image capturing instruction part **32** is operated by the technician when a still image of the site to be observed is captured. The treatment tool introduction port **33** is an opening through which a treatment tool is to be inserted into a treatment tool insertion path that is inserted inside the insertion section **20**. Note that illustration of the treatment tool insertion path and the treatment tool is omitted. A still image, assigned a reference sign **39**, is illustrated in FIG. **3**.

(35) The universal cord **22** is a connection cord that connects the endoscope **10** to the light source device **11**. The universal cord **22** includes therein a light guide **35**, a signal cable **36**, and a fluid tube (not illustrated), which are inserted inside the insertion section **20**.

(36) In addition, a tip part of the universal cord **22** includes a connector **37a** to be connected to the light source device **11** and a connector **37b** branching from the connector **37a** and to be connected to the processor device **12**.

(37) When the connector **37a** is connected to the light source device **11**, the light guide **35** and the fluid tube (not illustrated) are inserted into the light source device **11**. Consequently, necessary illumination light, water, and gas are supplied from the light source device **11** to the endoscope **10** through the light guide **35** and the fluid tube (not illustrated).

(38) As a result, the illumination light is radiated from an illumination window (not illustrated) of the tip surface **27a** of the tip part **27** toward the site to be observed. In addition, in response to an operation of pressing the air/water supply button **30**, gas or water is ejected from an air/water supply nozzle (not illustrated) of the tip surface **27a** of the tip part **27** toward the observation window (not illustrated) of the tip surface **27a**.

(39) When the connector **37b** is connected to the processor device **12**, the signal cable **36** and the processor device **12** are electrically connected to each other. Consequently, an imaging signal of the site to be observed is output from the imaging element **28** of the endoscope **10** to the processor device **12** through the signal cable **36**. Also, a control signal is output from the processor device **12** to the endoscope **10** through the signal cable **36**.

(40) In the present embodiments, the flexible endoscope is described as an example of the endoscope **10**. However, various types of electronic endoscopes capable of capturing a moving image of a site to be observed, such as a rigid endoscope, may be used as the endoscope **10**.

(41) The light source device **11** supplies illumination light to the light guide **35** of the endoscope **10** through the connector **37a**. White light or light in a specific wavelength range is usable as the illumination light. The illumination light may be a combination of white light and light in a specific wavelength range. The light source device **11** is configured to be able to appropriately select, as the illumination light, light in a wavelength range corresponding to an observation purpose.

(42) The white light may be light in a white wavelength range or light in a plurality of wavelength ranges. The specific wavelength range is a range narrower than the white wavelength range. As the light in the specific wavelength range, light in a single wavelength range may be used, or light in a plurality of wavelength ranges may be used. The light in the specific wavelength range may be referred to as special light.

(43) The processor device **12** controls the operation of the endoscope **10** through the connector **37b** and the signal cable **36**. The processor device **12** also acquires an imaging signal from the imaging element **28** of the endoscope **10** through the connector **37b** and the signal cable **36**. The processor device **12** uses a predetermined frame rate to acquire an imaging signal output from the endoscope **10**.

(44) The processor device **12** generates an endoscopic image, which is an observation image of the site to be observed, on the basis of the imaging signal acquired from the endoscope **10**. Herein, an endoscopic image **38** includes a moving image. The endoscopic image **38** may include the still

image **39**. Note that a moving image, assigned a reference sign **38a**, is illustrated in FIG. **3**. The endoscopic image **38** described in the embodiments is an example of a captured image.

(45) When the still image capturing instruction part **32** of the operation section **21** is operated, the processor device **12** generates the still image **39** of the site to be observed on the basis of the imaging signal acquired from the imaging element **28** in parallel with generation of the moving image. The still image **39** may be generated to have a resolution higher than the resolution of the moving image.

(46) When the endoscopic image **38** is generated, the processor device **12** performs image quality correction in which digital signal processing such as white balance adjustment and shading correction are used. The processor device **12** may add accessory information defined by the DICOM standard to the endoscopic image **38**. Note that DICOM is an abbreviation for Digital Imaging and Communications in Medicine.

(47) The endoscopic image **38** is an in-vivo image depicting the inside of a subject, that is, the inside of a living body. If the endoscopic image **38** is an image obtained through imaging using light in a specific wavelength range, the endoscopic image **38** is a special-light image. The processor device **12** then outputs the generated endoscopic image **38** to each of the display device **13** and the medical image processing apparatus **14**. The processor device **12** may output the endoscopic image **38** to a storage device (not illustrated) via a network (not illustrated) in accordance with a communication protocol compliant with the DICOM standard. Note that a network **140** illustrated in FIG. **2** may be used as the network.

(48) The display device **13** is connected to the processor device **12**. The display device **13** displays the endoscopic image **38** transmitted from the processor device **12**. The technician may perform an operation of moving the insertion section **20** forward and backward while checking the endoscopic image **38** displayed on the display device **13**. Upon detecting a lesion or the like at the site to be observed, the technician may operate the still image capturing instruction part **32** to capture a still image of the site to be observed.

(49) A computer is used as the medical image processing apparatus **14**. A keyboard, a mouse, and the like connectable to the computer are used as the input device **15**. The input device **15** and the computer may be connected to each other either with a cable or wirelessly. Various monitors connectable to the computer are used as the monitor device **16**.

(50) As the medical image processing apparatus **14**, a diagnosis assistant apparatus such as a workstation or a server apparatus may be used. In this case, the input device **15** and the monitor device **16** are provided for each of a plurality of terminals connected to the workstation or the like. Further, as the medical image processing apparatus **14**, a medical service assistant apparatus that assists creation of a medical report or the like may be used.

(51) The medical image processing apparatus **14** acquires the endoscopic image **38** and stores the endoscopic image **38**. The medical image processing apparatus **14** controls reproduction performed by the monitor device **16**. Note that the term “image” used herein includes a meaning of an electric signal representing the image and a meaning of image data such as information representing the image. The term “image” used herein means at least any of an image itself or image data.

(52) Further, the term “storing an image” can be interpreted as “saving an image”. “Storing an image” used herein means “storing an image in a non-transitory manner”. The medical image processing apparatus **14** may include a temporary storage memory that temporarily stores an image.

(53) The input device **15** is used to input an operation instruction for the medical image processing apparatus **14**. The monitor device **16** displays the endoscopic image **38** under the control of the medical image processing apparatus **14**. The monitor device **16** may function as a display device of various kinds of information in the medical image processing apparatus **14**.

(54) The medical image processing apparatus **14** may be connected to a storage device (not illustrated) via a network (not illustrated in FIG. **1**). The DICOM standard, a protocol compliant with the DICOM standard, and the like may be used as the image storage format and for the

communication between apparatuses via the network.

(55) As the storage device (not illustrated), a storage or the like that stores data in a non-transitory manner may be used. The storage device may be managed using a server apparatus (not illustrated). As the server apparatus, a computer that stores and manages various kinds of data may be used.

(56) A configuration including the medical image processing apparatus **14** and the monitor device **16** described in the embodiments is an example of a medical image processing system. Description of Medical Image Processing Apparatus According to First Embodiment

(57) Hardware Configuration of Medical Image Processing Apparatus

(58) FIG. 2 is a block diagram illustrating a hardware configuration of the medical image processing apparatus. The medical image processing apparatus **14** illustrated in FIG. 2 includes a control unit **120**, a memory **122**, a storage device **124**, a network controller **126**, a power supply device **128**, a display controller **130**, an input/output interface **132**, and an input controller **134**. Note that I/O illustrated in FIG. 2 represents the input/output interface **132**.

(59) The control unit **120**, the memory **122**, the storage device **124**, the network controller **126**, the display controller **130**, and the input/output interface **132** are connected to each other via a bus **136** so that data communication can be performed therebetween.

(60) Control Unit

(61) The control unit **120** functions as an overall control unit, various calculation units, and a storage control unit of the medical image processing apparatus **14**. The control unit **120** executes a program stored in a read-only memory (ROM) included in the memory **122**.

(62) The control unit **120** may download a program from an external storage device (not illustrated) via the network controller **126** and execute the downloaded program. The external storage device may be communicably connected to the medical image processing apparatus **14** via the network **140**.

(63) The control unit **120** uses, as a calculation area, a random access memory (RAM) included in the memory **122** and executes various processes in cooperation with various programs.

Consequently, various functions of the medical image processing apparatus **14** are implemented.

(64) The control unit **120** controls reading out of data from the storage device **124** and writing of data to the storage device **124**. The control unit **120** may acquire various kinds of data from an external storage device via the network controller **126**. The control unit **120** is capable of executing various processes such as calculations using the acquired various kinds of data.

(65) The control unit **120** may include one processor or two or more processors. Examples of the processor include a field programmable gate array (FPGA), a programmable logic device (PLD), and so on. An FPGA and a PLD are devices whose circuit configurations are changeable after being manufactured.

(66) Another example of the processor is an application-specific integrated circuit (ASIC). An ASIC includes a circuit configuration dedicatedly designed to execute specific processing.

(67) The control unit **120** may use two or more processors of the same kind. For example, the control unit **120** may use two or more FPGAs or two or more PLDs. The control unit **120** may use two or more processors of different kinds. For example, the control unit **120** may use one or more FPGAs and one or more ASICs.

(68) When the medical image processing apparatus **14** includes a plurality of control units **120**, the plurality of control units **120** may be configured using a single processor. As an example of configuring the plurality of control units **120** using a single processor, there is a form in which the single processor is configured using a combination of one or more central processing units (CPUs) and software and this processor functions as the plurality of control units **120**. Note that software used herein is synonymous with a program.

(69) As another example of configuring the plurality of control units **120** using a single processor, there is a form in which a processor that implements, with a single IC chip, the functions of the entire system including the plurality of control units **120**. Representative examples of the processor

that implements, with a single IC chip, the functions of the entire system including the plurality of control units **120** include a system on a chip (SoC). Note that IC is an abbreviation for integrated circuit.

(70) As described above, the control unit **120** is configured using one or more of various kinds of processors as the hardware structure.

(71) Memory

(72) The memory **122** includes a ROM (not illustrated) and a RAM (not illustrated). The ROM stores various programs to be executed in the medical image processing apparatus **14**. The ROM stores parameters, files, and the like used for executing various programs. The RAM functions as a temporary data storage area, a work area for the control unit **120**, and the like.

(73) Storage Device

(74) The storage device **124** stores various kinds of data in a non-transitory manner. The storage device **124** may be externally attached to the medical image processing apparatus **14**. Instead of or along with the storage device **124**, a large-capacity semiconductor memory device may be used.

(75) Network Controller

(76) The network controller **126** controls data communication between the medical image processing apparatus **14** and an external apparatus. The control of the data communication may include management of the traffic in the data communication. As the network **140** to which the medical image processing apparatus **14** is connected via the network controller **126**, a known network such as a local area network (LAN) may be used.

(77) Power Supply Device

(78) As the power supply device **128**, a large-capacity power supply device such as an uninterruptible power supply (UPS) is used. The power supply device **128** supplies power to each unit of the medical image processing apparatus **14** when the commercial power supply is cut off due to a power failure or the like.

(79) Display Controller

(80) The display controller **130** functions as a display driver that controls the monitor device **16** in accordance with a command signal transmitted from the control unit **120**.

(81) Input/Output Interface

(82) The input/output interface **132** communicably connects the medical image processing apparatus **14** and an external device to each other. A communication standard such as Universal Serial Bus (USB) may be used for the input/output interface **132**.

(83) Input Controller

(84) The input controller **134** converts the format of a signal input using the input device **15** into a format suitable for processing performed by the medical image processing apparatus **14**. Information input from the input device **15** via the input controller **134** is transmitted to each unit via the control unit **120**.

(85) Note that the hardware configuration of the medical image processing apparatus **14** illustrated in FIG. 2 is merely an example. Thus, addition, deletion, and modification may be appropriately made. Note that the hardware configuration of the medical image processing apparatus **14** illustrated in FIG. 2 is also applicable to embodiments other than a first embodiment.

(86) Description of Functional Blocks of Medical Image Processing Apparatus

(87) FIG. 3 is a functional block diagram of the medical image processing apparatus according to the first embodiment. The medical image processing apparatus **14** includes an image acquisition unit **40**, an image processing unit **42**, a display control unit **44**, and a storage unit **46**. The image acquisition unit **40** acquires the endoscopic image **38** from the processor device **12**. The image acquisition unit **40** stores the endoscopic image **38** in an endoscopic image storage unit **46a**.

(88) The image acquisition unit **40** may acquire the endoscopic image **38** from the processor device **12** via an information storage medium such as a memory card. The image acquisition unit **40** may acquire the endoscopic image **38** via the network **140** illustrated in FIG. 2.

(89) That is, the image acquisition unit **40** may acquire the moving image **38a** constituted by time-series frame images **38b**. The image acquisition unit **40** may acquire the still image **39** in the case where still image capturing is performed during capturing of the moving image **38a**.

(90) The image processing unit **42** includes a classification unit **48** and a region image generation unit **50**. The classification unit **48** performs automatic classification of a lesion from the endoscopic image **38**. The term “classification” used herein can be read as “discrimination”.

(91) Specifically, the classification unit **48** may classify the endoscopic image **38** into a predetermined class and derive a classification result. The classification unit **48** adds class information to the frame images **38b** constituting the moving image **38a** at the time of automatic classification of the endoscopic image **38**. The classification unit **48** may add the class information to all the frame images **38b**, or may add the class information to the frame images **38b** of every several frames. The classification unit **48** may add the class information to the still image **39**. The class and the class information can be read as a classification result.

(92) The classification unit **48** stores the classification result in a classification storage unit **46b** in association with the frame images **38b**. Table 1 below illustrates examples of classes used by the classification unit **48**.

(93) TABLE-US-00001 TABLE 1 Class Specific example Tumor or non-tumor Tumor, Non-tumor Classification of NICE classification, JNET endoscopic findings classification, etc. Type Hyperplastic polyp, Adenoma, Intramucosal carcinoma, Highly invasive carcinoma, Inflammatory polyp, etc. Note that NICE in Table 1 above is an abbreviation for NBI International Colorectal Endoscopic Classification. NBI is an abbreviation for Narrow Band Imaging. JNET is an abbreviation for The Japan NBI Expert Team.

(94) The region image generation unit **50** generates a region image representing a region that contributes to classification in the endoscopic image **38**. The region image generation unit **50** stores the region image in a region image storage unit **46c**. The region image, assigned a reference sign **208**, is illustrated in FIG. 4.

(95) The display control unit **44** transmits, to the monitor device **16**, a display control signal that causes the monitor device **16** to display the endoscopic image **38** and the region image generated using the region image generation unit **50**. The display control unit **44** transmits, to the monitor device **16**, a display control signal that causes the monitor device **16** to display text information representing the classification result of the endoscopic image **38** derived using the classification unit **48**. The display control unit **44** described in the embodiments is an example of a display signal transmission unit.

(96) The monitor device **16** displays the endoscopic image **38**, the region image, and the text information representing the classification result in the same screen. The monitor device **16** displays the endoscopic image **38** and the region image in different regions in the screen. Text information, assigned a reference sign **212**, representing the classification result is illustrated in FIG. 4. Details of the screen displayed using the monitor device **16** will be described later.

(97) The storage unit **46** includes the endoscopic image storage unit **46a**, the classification storage unit **46b**, and the region image storage unit **46c**. The endoscopic image storage unit **46a** stores the endoscopic image **38** acquired using the image acquisition unit **40**.

(98) The image processing unit **42** reads out the endoscopic image **38** stored in the endoscopic image storage unit **46a** and performs image processing on the endoscopic image **38**. The display control unit **44** reads out the endoscopic image **38** stored in the endoscopic image storage unit **46a** and causes the monitor device **16** to display the endoscopic image **38**.

(99) The classification storage unit **46b** stores the class of the endoscopic image **38** classified using the classification unit **48** in association with the endoscopic image **38**. Specifically, the classification storage unit **46b** stores, in association with the frame image **38b**, the class of the frame image **38b** constituting the moving image **38a**. The display control unit **44** reads out the classification result from the classification storage unit **46b** and causes the monitor device **16** to

display the text information or the like representing the classification result.

(100) The region image storage unit **46c** stores the region image generated using the region image generation unit **50**. The display control unit **44** reads out the region image from the region image storage unit **46c** and causes the monitor device **16** to display the region image.

(101) One or more storage elements may be used as the storage unit **46** illustrated in FIG. **3**. That is, the storage unit **46** may include three storage elements corresponding to the endoscopic image storage unit **46a**, the classification storage unit **46b**, and the region image storage unit **46c**, respectively. A plurality of storage elements may be used as each of the endoscopic image storage unit **46a**, the classification storage unit **46b**, and the region image storage unit **46c**. Furthermore, two or all of the endoscopic image storage unit **46a**, the classification storage unit **46b**, and the region image storage unit **46c** may be configured using a single storage element. Description of Display Screen Displayed on Monitor Device

(102) Description of Region Image

(103) FIG. **4** is an explanatory diagram of a display screen used in the first embodiment. A display screen **200** illustrated in FIG. **4** includes an endoscopic image display area **202**, a region image display area **204**, and a classification result display area **206**.

(104) The endoscopic image display area **202** is an area in which the endoscopic image **38** is displayed. The still image **39** may be displayed in the endoscopic image display area **202**. The endoscopic image **38** and the still image **39** may be displayed in a switching manner in the endoscopic image display area **202**. A reference sign **209** denotes a classification contribution region **209** which is a region that contributes to classification of the endoscopic image **38**. FIG. **4** schematically illustrates the classification contribution region **209**.

(105) The region image **208** is displayed in the region image display area **204**. In the region image **208**, a classification contribution corresponding region **210** is displayed with highlight. The classification contribution corresponding region **210** is a region, in the region image **208**, corresponding to the classification contribution region **209**.

(106) The text information **212** representing the classification result is displayed in the classification result display area **206**. FIG. **4** illustrates an example in which “neoplastic”, which is the English notation for a tumor, is displayed as the text information **212** in the classification result display area **206**. Note that the text information **212** may be in a language of any kind. That is, Japanese notation or foreign language notation other than English may be used for the text information **212**. An abbreviation may also be used for the text information **212**.

(107) FIG. **5** is an explanatory diagram of a display screen according to a comparative example. A comparative screen **220** displays the result of automatic classification, and indicates an example in which a tumor **222** is found from the endoscopic image **38** and the endoscopic image **38** is classified to the tumor. The endoscopic image **38** and the text information **212** representing the classification result are displayed in the comparative screen **220**. Note that FIG. **5** schematically illustrates the tumor **222**.

(108) The text information **212** representing the classification result is displayed in the comparative screen **220** illustrated in FIG. **5**. However, when the endoscopic image **38** is difficult to classify, an incorrect classification result may be output. On the other hand, the region image **208** corresponding to the endoscopic image **38** is displayed in the display screen **200** illustrated in FIG. **4**, and the classification contribution corresponding region **210** is displayed in the region image **208**. This allows an observer to visually grasp which region in the endoscopic image **38** the classification unit **48** performs classification on the basis of. In addition, the region image **208** may serve as an index of the reliability of the classification result.

(109) FIG. **6** is an explanatory diagram of a display screen displayed when the reliability of the classification result is low. A display screen **200a** illustrated in FIG. **6** corresponds to an example of the case where the classification result is incorrect and the classification is performed on the basis of a region different from the tumor **222** in the endoscopic image **38**.

(110) In a region image **208a** illustrated in FIG. 6, a region **230** different from a region to be set as a classification contribution corresponding region **210a** is displayed as the classification contribution corresponding region.

(111) FIG. 7 is an explanatory diagram of a display screen of another example displayed when the reliability of the classification result is low. A display screen **200b** illustrated in FIG. 7 corresponds to an example displayed in the case where the tumor **222** in the endoscopic image **38** fails to be found.

(112) In a region image **208b** illustrated in FIG. 7, a region **210b** to be set as the classification contribution corresponding region is not displayed. When the display screen **200b** illustrated in FIG. 7 is displayed, it is considered that the classification unit **48** fails to find a target to be classified due to reasons such as the tumor **222** being blurred and the size of the tumor **222** being small.

(113) In such a case, the operator operates the endoscope **10** illustrated in FIG. 1 to adjust the focus or to display the lesion and a peripheral region of the lesion in an enlarged manner. This allows the classification unit **48** to perform classification correctly. That is, displaying the region image **208** together with the endoscopic image **38** may serve as a suggestion that prompts the operator to perform an operation that causes the classification unit **48** to derive the correct classification result.

(114) In the present embodiment, the example in which the endoscopic image display area **202** and the region image display area **204** are displayed in the display screen **200** displayed on the single monitor device **16** has been described. Alternatively, the display screen **200** including the endoscopic image display area **202** may be displayed on one of two monitor devices, and the display screen **200** including the region image display area **204** may be displayed on the other monitor device.

(115) In addition, the endoscopic image display area **202** and the region image display area **204** may be displayed in the single display screen **200** displayed on the single monitor device **16** so as to be switched in a time division manner. Further, the region image display area **204** may be displayed to be superimposed on the endoscopic image display area **202**. For example, the region image display area **204** may be displayed to be superimposed, at a location where observation of the endoscopic image **38** is not hindered in the endoscopic image display area **202**, such as a lower left corner of the endoscopic image display area **202** illustrated in FIG. 7.

(116) The observer tends to dislike movement of the viewpoint because the observer observes the endoscopic image while performing a precise operation. When the region image display area **204** is displayed to be superimposed on the endoscopic image display area **202**, the endoscopic image display area **202** and the region image display area **204** in the display screen **200b** are arranged at closer locations. This effectively reduces movement of the viewpoint of the observer.

(117) FIG. 8 is an explanatory diagram of a display screen displayed when a plurality of lesions are present. In a display screen **200d** illustrated in FIG. 8, a first classification contribution region **209a** and a second classification contribution region **209b** are extracted for the plurality of lesions that are present in the endoscopic image **38**.

(118) In a region image **208g**, a first classification contribution corresponding region **210g** corresponding to the first classification contribution region **209a** and a second classification contribution corresponding region **210h** corresponding to the second classification contribution region **209b** are displayed. A display manner is used that enables the first classification contribution corresponding region **210g** and the second classification contribution corresponding region **210h** to be distinguished from each other.

(119) Further, in a classification result display area **206a** of the display screen **200d**, first text information **212e** representing a first classification result for the first classification contribution corresponding region **210g** and second text information **212f** representing a second classification result for the second classification contribution corresponding region **210h** are displayed.

(120) When lesions of different classifications, such as a non-neoplastic lesion and a neoplastic

lesion, are present, it is difficult for a system that outputs one classification result from the endoscopic image **38** to return an appropriate classification result. In contrast, the medical image processing apparatus **14** according to the present embodiment uses the display manners for the respective classifications in the region image **208g** as illustrated in FIG. **8** to depict the first classification contribution region **209a** and the second classification contribution region **209b**. This enables appropriate classification results to be obtained even when a plurality of lesions are present in the endoscopic image **38** and the classifications of the plurality of lesions are different from each other.

(121) Description of Display of Classification Result

(122) FIG. **9** is an explanatory diagram of a display example of a classification result. FIG. **9** illustrates an example in which a specific class is displayed as a classification result. Text information **212a** illustrated in FIG. **9** indicates that the classification result is NICE 1.

(123) FIG. **10** is an explanatory diagram of an example of displaying membership probabilities as the classification results. FIG. **10** illustrates an example in which the membership probabilities for the respective classes are displayed as the classification results. Text information **212b** illustrated in FIG. **10** indicates that the membership probability for NICE 1 is 98%, the membership probability for NICE 2 is 2%, and the membership probability for NICE 3 is 0%.

(124) The text information **212b** illustrated in FIG. **10** may be text indicating that the membership probability for NICE 1 is 98% alone. The text information **212b** illustrated in FIG. **10** may be text indicating that the membership probability for NICE 1 is 98% and the membership probability for NICE 2 is 2%.

(125) FIG. **11** is an explanatory diagram of an example of displaying scores as the classification results. FIG. **11** illustrates an example in which the scores for the respective classes are displayed as the classification results. Text information **212c** illustrated in FIG. **11** indicates that the score for NICE 1 is 1.23, the score for NICE 2 is 0.002, and the score for NICE 3 is 0.05. The membership probabilities illustrated in FIG. **10** and the scores illustrated in FIG. **11** are examples of membership degrees for classes.

(126) Procedure of Medical Image Processing Method

(127) FIG. **12** is a flowchart illustrating a procedure of a medical image processing method. In an endoscopic image acquisition step **S10**, the image acquisition unit **40** illustrated in FIG. **3** acquires the endoscopic image **38**. In an endoscopic image storage step **S12**, the image acquisition unit **40** stores the endoscopic image **38** acquired in the endoscopic image acquisition step **S10** in the endoscopic image storage unit **46a**.

(128) In a classification step **S14**, the classification unit **48** classifies the endoscopic image **38** into a predetermined class. In a classification result storage step **S16**, the classification unit **48** stores the classification result derived in the classification step **S14** in the classification storage unit **46b**.

(129) In a region image generation step **S18**, the region image generation unit **50** generates a region image such as the region image **208** illustrated in FIG. **4** on the basis of the classification result. In a region image storage step **S20**, the region image generation unit **50** stores the region image generated in the region image generation step **S18** in the region image storage unit **46c**.

(130) In a display signal transmission step **S22**, the display control unit **44** transmits a display signal to the monitor device **16**. The display signal transmitted from the display control unit **44** to the monitor device **16** includes a display signal representing the endoscopic image **38** and the region image **208**. The display signal transmitted from the display control unit **44** to the monitor device **16** may include a display signal representing the classification result.

(131) Modifications of Display Screen

(132) Modification of Region Image

(133) FIG. **13** is an explanatory diagram of a display screen according to a modification of a region image. In a display screen **200c** illustrated in FIG. **13**, a reduced-size image of the endoscopic image **38** is combined in the background of a region image **208c**. That is, in the region image **208c**,

the classification contribution corresponding region **210** is displayed to be superimposed on the reduced-size endoscopic image **38**. The reduced-size image of the endoscopic image **38** may have a lower resolution than the endoscopic image **38**.

(134) Modifications of Classification Result

(135) FIG. **14** is an explanatory diagram of a display screen according to a first modification of the classification result. In a region image **208d** illustrated in FIG. **14**, text information **212d** representing the classification result is displayed to be superimposed. In the region image **208d**, emphasis on the text information **212d** such as changing the color of the text information **212d** from the color of the classification contribution corresponding region **210** may be used.

(136) FIG. **15** is an explanatory diagram of a display screen according to a second modification of the classification result. In a display screen **200e** illustrated in FIG. **15**, a frame **201** of the endoscopic image **38** and a frame **207** of a region image **208h** are colored in accordance with the classification result. Note that the frame **201** alone may be colored, or the frame **207** alone may be colored. That is, in the display screen **200e**, at least any of the frame **201** of the endoscopic image **38** or the frame **207** of the region image **208h** is colored in accordance with the classification result.

(137) Since the observer observes the endoscopic image while performing a precise operation, there is a concern that the movement of the viewpoint of the observer, the visual recognition of the text information by the observer, and the like may adversely affect the operation performed by the observer. In contrast, the operator who views the display screen **200e** illustrated in FIG. **15** may grasp the classification result with almost no movement of the viewpoint.

(138) In addition to coloring the frame **201** or the like in accordance with the classification result, the medical image processing apparatus **14** may be configured such that the frame **201** or the like is colored in the case where the classification result is a specific classification such as a tumor and the frame **201** or the like is not colored in the case where the classification result is another classification. Alternatively, the medical image processing apparatus **14** may color the frame **201** or the like in the case of exceptional determination (described later), or may change the color in accordance with the degree of reliability in the case where the display is changed in accordance with the degree of reliability.

(139) The configuration of changing the color may include a configuration of changing the density of the same color. For example, deep red may be used if the endoscopic image **38** is classified into a tumor, and light red may be used if the endoscopic image **38** is classified into a non-tumor. An object to be colored is not limited to the frame **201** or the like. A region other than the frames **201** and **207** may be colored. Furthermore, a non-color-based display manner such a display manner using a symbol can be used as long as the display manner makes the movement of the viewpoint of the operator less and makes it easier for the operator to grasp the classification result than the text information.

(140) Other Modifications

(141) The display manner of the region image may be changed in accordance with the classification result. For example, in a case where the endoscopic image **38** is classified into two classes of a tumor and a non-tumor, the color used in the case of the tumor may be changed from the color used in the case of the non-tumor. In such an example, red may be used if the endoscopic image **38** is classified into the tumor, and blue may be used if the endoscopic image **38** is classified into the non-tumor.

(142) Alternatively, the display screen may be configured such that the plurality of region image display areas **204** are displayable, and the location of the region image **208** may be changed in accordance with the classification result. For example, in a display screen in which two region image display areas **204** vertically displayable, the upper region image display area **204** may be used if the endoscopic image **38** is classified into the tumor, and the lower region image display area **204** may be used if the endoscopic image **38** is classified into the non-tumor. The modifications of the display screen described above may assist the visibility of the operator. The

display manner according to the classification result described in the embodiments is an example of a depicting manner.

(143) Display Examples of Region Image According to Degree of Contribution to Classification

(144) FIG. 16 is an explanatory diagram of a region image to which density according to a degree of contribution is applied. In the display screen 200e illustrated in FIG. 16, the density according to the degree of contribution is applied to a classification contribution corresponding region 210e in a region image 208c.

(145) In the classification contribution corresponding region 210e illustrated in FIG. 16, a central portion 211a is colored deeper and a peripheral portion 211b is colored lighter. This indicates that the central portion 211a has a high degree of contribution, and the peripheral portion 211b has a low degree of contribution. Note that the degree of contribution may be classified in there or more levels.

(146) FIG. 17 is an explanatory diagram of a region image to which a heat map according to the degree of contribution is applied. In the display screen 200f illustrated in FIG. 17, a heat map according to the degree of contribution is applied to a classification contribution corresponding region 210f in a region image 208f.

(147) In the classification contribution corresponding region 210f illustrated in FIG. 17, red is used for the central portion 211a and blue is used for the peripheral portion 211b. An intermediate color that changes from red to blue is used for an intermediate region 211c between the central portion 211a and the peripheral portion 211b. Examples of the intermediate color include orange, yellow, green, and so on.

(148) The classification contribution corresponding region 210f illustrated in FIG. 17 indicates that the degree of contribution of the central portion 211a is high, the degree of contribution decreases from the central portion 211a toward the peripheral portion 211b, and the degree of contribution of the peripheral portion 211b is the lowest. Note that two or more of the modifications described herein may be appropriately combined with one another.

(149) Detailed Description of Classification and Depiction of Region that Contributes to Classification

(150) A specific example of classification of the endoscopic image 38 and depiction of a region that contributes to the classification will be described next.

(151) Pattern 1

(152) As a pattern 1, an example is presented in which a feature quantity is calculated from the endoscopic image 38, classification is performed on the basis of the feature quantity, and a region that contributes to the classification is depicted. A method based on a support vector machine (SVM) or the like may be used in the classification based on the feature quantity. For example, a blood vessel region is extracted from the endoscopic image 38, and a feature quantity of the extracted blood vessel region is calculated.

(153) Other methods include a method of performing texture analysis on the endoscopic image 38 and calculating a feature quantity using the analysis result, a method of calculating a local feature quantity such as scale-invariant feature transform (SIFT), and so on.

(154) The feature quantity calculated using any of the aforementioned methods is analyzed in units of regions obtained in the case where the target image is divided into a plurality of regions. In this way, the class membership probability can be calculated in units of regions. This consequently enables depiction processing to be performed for individual regions based on the respective class membership probabilities in units of regions. The plurality of regions described in the embodiments are an example of a plurality of regions set in a captured image.

(155) Pattern 2

(156) As a pattern 2, an example is presented in which information of an intermediate layer of a convolutional neural network is analyzed to classify the endoscopic image 38 and a region that contributes to the classification is depicted. Such a method enables classification and depiction to

be processed in parallel.

(157) FIG. **18** is a block diagram illustrating an example of a configuration of a classification unit that employs a convolutional neural network. Hereinafter, the convolutional neural network may be abbreviated as CNN. The classification unit that employs the convolutional neural network described in the embodiments is an example of a deep learning device.

(158) A classification unit **300** illustrated in FIG. **18** is an example of the classification unit **48** illustrated in FIG. **3**. The classification unit **300** includes a CNN **302**, an error calculation unit **304**, and a parameter updating unit **306**.

(159) The CNN **302** performs image recognition on the type of a lesion contained in the endoscopic image **38**. The CNN **302** has a structure of a plurality of layers and holds a plurality of weight parameters. The weight parameters of the CNN **302** are updated from the initial values to the optimum values. This may change an untrained model to a trained model.

(160) The CNN **302** includes an input layer **310**, an intermediate layer **312**, and an output layer **314**. The intermediate layer **312** includes a plurality of sets of a convolutional layer **320** and a pooling layer **322**. The intermediate layer **312** also includes a fully connected layer **324**. Each layer is structured such that a plurality of nodes are connected to one another by edges.

(161) The endoscopic image **38** serving as a target to be learned is input to the input layer **310**. The intermediate layer **312** extracts features from the endoscopic image **38** input thereto from the input layer **310**. The convolutional layer **320** performs filtering processing on nearby nodes in the previous layer to acquire a feature map. Note that the filtering processing is synonymous with a convolution operation using a filter.

(162) The pooling layer **322** reduces the size of the feature map output from the convolutional layer **320** to generate a new feature map. The convolutional layer **320** plays a role of extracting features, such as extracting edges, from the endoscopic image **38**.

(163) The pooling layer **322** plays a role of providing robustness so that the extracted features are not affected by translation or the like. Note that the intermediate layer **312** is not limited to the case where the convolutional layer **320** and the pooling layer **322** constitute a single set, and the case where the convolutional layers **320** are consecutively arranged and a configuration including a normalization layer (not illustrated) are also possible.

(164) The output layer **314** classifies the type of the lesion contained in the endoscopic image **38** on the basis of the features extracted using the intermediate layers **312**. The trained CNN **302** may classify, for example, a medical image into two classes, that is, a tumor or a non-tumor. The recognition result may be obtained as two kinds of scores corresponding to the tumor or the non-tumor.

(165) Any initial values are set for the coefficient and the offset value of the filter used in the convolutional layer **320** and the connection weight to the next layer of the fully connected layer **324** in the CNN **302** that has not been trained yet.

(166) The error calculation unit **304** acquires the recognition result output from the output layer **314** of the CNN **302** and correct answer data **370** for the endoscopic image **38**, and calculates an error therebetween. Examples of the error calculation method include the softmax cross entropy, the sigmoid, and the like.

(167) The parameter updating unit **306** adjusts the weight parameters of the CNN **302** by applying backpropagation based on the error calculated using the error calculation unit **304**. The parameter updating unit **306** repeatedly performs the parameter adjustment processing and repeatedly performs training until the difference between the output of the CNN **302** and the correct answer data **370** becomes small.

(168) The classification unit **300** performs training for optimizing each parameter of the CNN **302** by using a data set of the endoscopic images **38** stored in a database (not illustrated) to generate a trained model.

(169) FIG. **19** is a conceptual diagram of the shape transition of the feature quantity in the

convolutional neural network. A reference sign **330** denotes schematized information. Input data represents information **330a** of the input layer **310**. Conv.1 to Conv.5 represent the intermediate layer **312**. FC6 to FC8 represent the fully connected layer **324**. In FIG. **19**, an information amount of the information **330a** of the input layer **310** and an information amount of information **330b** of the intermediate layer **312** are represented as width×height×number of channels.

(170) The convolutional neural network performs final output while aggregating spatial information of an input image. As illustrated in FIG. **19**, the convolutional neural network aggregates spatial information having a width and a height as the processing advances from the input layer **310** to the intermediate layer **312**. Consequently, the spatial information is lost in information **330c** of the fully connected layer **324** and thereafter.

(171) That is, analysis of the information **330b** of the intermediate layer **312** enables a feature retaining spatial information to be extracted. However, in general, the number of channels of the information **330b** of the intermediate layer **312** is larger than that of the information **330a** of the input layer **310**.

(172) In the example illustrated in FIG. **19**, the number of channels of the information **330a** is three in the input layer **310**, whereas the number of channels of the information **330b** is increased to **384** in Conv. 4 of the intermediate layer **312**. Accordingly, how to aggregate the information in the channel direction to depict the classification contribution region becomes an issue. Examples in which information in the channel direction is aggregated to depict the classification contribution region will be described below.

First Example

(173) In general, a channel that greatly contributes to the final output tends to have a large absolute value. Accordingly, the absolute values of information in the respective channels of the intermediate layer **312** are compared with each other, and a channel having a larger absolute value is extracted. The extracted channel is depicted. This enables a region that contributes more to the final output to be depicted.

(174) When the channel is extracted, a plurality of channels may be extracted. As an example of the case where a plurality of channels are extracted, there is an example in which a predetermined number of channels are extracted in descending order of the absolute values. When a plurality of channels are extracted, the plurality of channels may be averaged.

Second Example

(175) Principal component analysis may be performed in the channel direction of the intermediate layers **312** to extract a principal component, and the extracted principal component may be depicted. For example, the extracted principal component can be depicted by reducing the dimensions of the channels to one dimension. In the second example, since information of all channels is depicted, more accurate depiction than that in the first example is possible.

Third Example

(176) The final output result is a score for each class label of the classification. The degree of contribution to the score of each class label of the classification can be derived using differentiation. For example, Gradient-weighted Class Activation Mapping (Grad-CAM) may be used to derive the degree of contribution to the score of each class label of the classification.

(177) Let $y_{sup.c}$ denote the score of an arbitrary class c . Let $A_{sup.k}$ denote a feature map of a k -th channel of an arbitrary intermediate layer. Let $A_{sup.k.sub.ij}$ denote the values of the coordinates (i,j) of the feature map $A_{sup.k}$. A map $L_{sup.C.sub.Grad-CAM}$ in which the degree of contribution of the class c is depicted is obtained by Equation below.

$$(178) L_{Grad-CAM}^c = \text{ReLU} \left(\frac{\partial y_{sup.c}}{\partial A_{sup.k}} \right) A_{sup.k} = \frac{1}{Z} \cdot \frac{\partial y_{sup.c}}{\partial A_{sup.k}} \cdot Z = \frac{1}{Z} \cdot \frac{\partial y_{sup.c}}{\partial A_{sup.k}} \cdot Z$$

$$\text{ReLU}(x) = \max\{x, 0\}$$

(179) The map $L_{sup.C.sub.Grad-CAM}$ represents a region image depicting the classification contribution region in which information in the channel direction is aggregated.

(180) FIG. 20 is a conceptual diagram of depiction based on information of an intermediate layer of a convolutional neural network. The example illustrated in FIG. 20 presents an example in which an image to be processed 380 including two kinds of animals is classified into one animal 381a or another animal 381b and a region serving as a basis of the classification is depicted.

(181) A first classified image 382 presents an example in which the image to be processed 380 is classified into the one animal 381a and a region 384 serving as the basis of the classification is depicted. A second classified image 386 presents an example in which the image to be processed 380 is classified into the other animal 381b and a region 388 serving as the basis of the classification is depicted.

(182) When the pattern 2 is used in classification of the endoscopic image 38, the image to be processed 380 may be replaced with the endoscopic image 38, and the two kinds of animals may be replaced with two kinds of classes. The two kinds of animals may be replaced with two kinds of feature regions.

(183) Pattern 3

(184) As a pattern 3, an example is presented in which a region is depicted using a segmentation technique. FIG. 21 is an explanatory diagram of the segmentation technique. FIG. 21 illustrates an example in which an image to be classified 390 including a person 391a and an animal 391b is classified into a person region 394, an animal region 396, and a background region 398. A reference sign 392 denotes a classified image.

(185) With the segmentation technique, a class membership probability is obtained for each region. Each region can be depicted based on the membership probability. Classification can be performed on the image to be classified 390 on the basis of the regions obtained using the segmentation technique. For example, a class having the largest area in the image to be classified 390 among the classes used in classification may be set as the classification result.

Advantageous Effects of First Embodiment

(186) With the medical image processing apparatus according to the first embodiment, the following advantageous effects can be obtained. [1] The endoscopic image 38 is displayed in the endoscopic image display area 202 of the display screen 200 displayed on the monitor device 16. The region image 208 is displayed in the region image display area 204 different from the endoscopic image display area 202 of the display screen 200. The region image 208 includes the classification contribution corresponding region 210 corresponding to the classification contribution region 209 that contributes to classification of the endoscopic image 38. This allows an observer to visually grasp which region in the endoscopic image 38 the classification unit 48 performs classification on the basis of. In addition, the region image 208 may serve as an index of the reliability of the classification result. [2] The classification result is displayed on the monitor device 16. This allows the observer to visually recognize the classification result of the endoscopic image 38. [3] A membership degree for each class is derived as the classification result. The membership degree of each class is displayed on the monitor device 16. As the membership degree, a membership probability for each class or a score for each class may be used. This allows the observer to visually recognize the membership degree for each class. [4] A display manner of the region image 208 may be changed in accordance with the classification result. This may improve the visibility of the classification result.

Description of Medical Image Processing Apparatus According to Second Embodiment

(187) A medical image processing apparatus according to a second embodiment will be described next. In the medical image processing apparatus according to the second embodiment, exception determination is added to classification of the endoscopic image 38. That is, a lesion region of the endoscopic image 38 is depicted using the pattern 1, the pattern 2, and the pattern 3 described in the first embodiment, and then the region image is analyzed to make exception determination.

(188) For example, if the area of the lesion region detected from the endoscopic image 38 is equal to or less than a predetermined threshold value, the classification result may be set as undetectable.

If there are a plurality of lesion regions detected from the endoscopic image **38** and the areas of all the lesion regions are equal to or greater than a predetermined threshold value, the classification result may be set as undetectable.

(189) Note that the medical image processing apparatus according to the second embodiment employs substantially the same hardware and functional blocks as those of the medical image processing apparatus **14** according to the first embodiment. Thus, description of the hardware and functional blocks of the medical image processing apparatus according to the second embodiment will be omitted.

(190) Display Screen Displayed in Case of being Undetectable

(191) FIG. **22** is an explanatory diagram of a display screen indicating the classification result of “undetectable”. In a display screen **400** illustrated in FIG. **22**, undetectable indicating the classification result of “undetectable” is displayed as text information **412**. When the text information **412** representing undetectable is displayed, a classification contribution corresponding region **410** is not displayed in a region image **408**. Note that FIG. **22** schematically illustrates, using a solid line, a lesion region **407** that is undetectable. The lesion region **407** that is detectable is schematically illustrated using a two-dot chain line.

(192) A configuration in which the text information representing the classification result is overwritten with text information representing undetectable may be used as the text information **412**. The display screen **400** may employ a configuration indicating that the lesion is undetectable, separately from the text information indicating the classification result.

(193) The classification unit **48** illustrated in FIG. **3** may analyze the region image **408** to quantify the degree of reliability of the classification. The display control unit **44** may display a numerical value such as a score representing the degree of reliability of the classification in the display screen **400**. Note that a reference sign **402** illustrated in FIG. **22** denotes the endoscopic image display area. A reference sign **404** denotes the region image display area. A reference sign **406** denotes the classification result display area.

(194) Display Screen Displayed in Case of being Undeterminable

(195) FIG. **23** is an explanatory diagram of a display screen indicating the classification result of “undeterminable”. In a display screen **400a** illustrated in FIG. **23**, undeterminable indicating the classification result of “undeterminable” is displayed as text information **412a**. When the text information **412a** representing undeterminable is displayed, a classification contribution corresponding region **410a** indicating the undeterminable state is displayed in a region image **408a**.

(196) Examples of the classification contribution corresponding region **410a** indicating the undeterminable state include an example in which two types of display manners corresponding to different classifications coexist. FIG. **23** illustrates the classification contribution corresponding region **410a** in which two colors coexist in the case where the color is changed in accordance with the classification. Note that a reference sign **409** in FIG. **23** denotes the classification contribution region.

Advantageous Effects of Second Embodiment

(197) With the medical image processing apparatus according to the second embodiment, the following advantageous effects can be obtained. [1] When a lesion in the endoscopic image **38** is undetectable, the text information **412** indicating that the lesion is undetectable is displayed. When a lesion in the endoscopic image **38** is undeterminable, the text information **412** indicating that the lesion is undeterminable is displayed. This allows the operator to grasp the inappropriate classification. [2] The degree of reliability of classification is calculated on the basis of the region image. The degree of reliability is displayed in the display screen. This enables the degree of reliability of classification to be recognized.

Description of Medical Image Processing Apparatus According to Third Embodiment

(198) FIG. **24** is a functional block diagram of a medical image processing apparatus according to a third embodiment. A medical image processing apparatus **14a** illustrated in FIG. **24** includes a

storage instruction acquisition unit **41**, compared to the medical image processing apparatus **14** illustrated in FIG. **3**. The storage instruction acquisition unit **41** acquires an instruction to store the endoscopic image **38**, transmitted from the processor device **12** illustrated in FIG. **1**. The storage instruction acquisition unit **41** may acquire the instruction to store the endoscopic image **38** from the endoscope **10**.

(199) In response to the storage instruction acquisition unit **41** acquiring the instruction to store the endoscopic image **38**, the endoscopic image **38** acquired using the image acquisition unit **40** is stored in the endoscopic image storage unit **46a** in association with at least any of the classification result, the region image, the exception determination result, or the degree of reliability.

(200) The medical image processing apparatus **14a** may combine the classification result and the like with the endoscopic image **38** and store the combined result, or may separately store the classification result and the like and the endoscopic image **38**. In response to the operator operating an operation button (not illustrated) or the like, the instruction to store the endoscopic image **38** may be transmitted from the processor device **12**. Alternatively, the instruction to store the endoscopic image **38** may be automatically transmitted from the processor device **12** on the basis of the classification result or the like. When the degree of reliability is equal to or higher than a predetermined threshold value, the processor device **12** may determine that the medical image processing apparatus **14a** has obtained an appropriate classification result and automatically transmit the instruction to store the endoscopic image **38**.

Advantageous Effects of Third Embodiment

(201) With the medical image processing apparatus according to the third embodiment, the following advantageous effects can be obtained. [1] When a user creates a report after an endoscopic examination, the user may check whether the classification result or the like is appropriate. [2] At the time of re-examination, examination results of the preceding examinations may be referred to.

Modifications of Endoscope System

Modification of Processor Device

(202) The processor device **12** may have the functions of the medical image processing apparatus **14**. That is, the processor device **12** and the medical image processing apparatus **14** may be integrated together. In such an embodiment, the display device **13** may also serve as the monitor device **16**. The processor device **12** may include a connection terminal to which the input device **15** is connected.

(203) Modification of Illumination Light

(204) One example of the medical image acquirable using the endoscope system **9** according to the present embodiments is a normal-light image acquired by radiating light in a white range or light in a plurality of wavelength ranges as the light in the white range.

(205) Another example of the medical image acquirable using the endoscope system **9** according to the present embodiments is an image acquired by radiating light in a specific wavelength range. A range narrower than the white range may be used as the specific wavelength range. The following modifications may be employed.

First Example

(206) A first example of the specific wavelength range is a blue range or a green range in a visible range. The wavelength range of the first example includes a wavelength range of 390 nm or more and 450 nm or less or a wavelength range of 530 nm or more and 550 nm or less, and the light of the first example has a peak wavelength in the wavelength range of 390 nm or more and 450 nm or less or the wavelength range of 530 nm or more and 550 nm or less.

Second Example

(207) A second example of the specific wavelength range is a red range in the visible range. The wavelength range of the second example includes a wavelength range of 585 nm or more and 615 nm or less or a wavelength range of 610 nm or more and 730 nm or less, and the light of the second

example has a peak wavelength in the wavelength range of 585 nm or more and 615 nm or less or the wavelength range of 610 nm or more and 730 nm or less.

Third Example

(208) A third example of the specific wavelength range includes a wavelength range in which an absorption coefficient is different between oxyhemoglobin and deoxyhemoglobin, and the light of the third example has a peak wavelength in the wavelength range in which the absorption coefficient is different between oxyhemoglobin and deoxyhemoglobin. The wavelength range of this third example includes a wavelength range of 400 ± 10 nm, a wavelength range of 440 ± 10 nm, a wavelength range of 470 ± 10 nm, or a wavelength range of 600 nm or more and 750 nm or less, and the light of the third example has a peak wavelength in the wavelength range of 400 ± 10 nm, the wavelength range of 440 ± 10 nm, the wavelength range of 470 ± 10 nm, or the wavelength range of 600 nm or more and 750 nm or less.

Fourth Example

(209) A fourth example of the specific wavelength range is a wavelength range of excitation light that is used to observe fluorescence emitted by a fluorescent substance in a living body and excites this fluorescent substance. For example, the specific wavelength range of the fourth example is a wavelength range of 390 nm or more and 470 nm or less. Note that observation of fluorescence may be referred to as fluorescence observation.

Fifth Example

(210) A fifth example of the specific wavelength range is a wavelength range of infrared light. The wavelength range of this fifth example includes a wavelength range of 790 nm or more and 820 nm or less or a wavelength range of 905 nm or more and 970 nm or less, and the light of the fifth example has a peak wavelength in the wavelength range of 790 nm or more and 820 nm or less or the wavelength range of 905 nm or more and 970 nm or less.

(211) Generation Example of Special-Light Image

(212) The processor device **12** may generate a special-light image having information in the specific wavelength range on the basis of a normal-light image obtained through imaging using white light. Note that the term “generation” used herein includes “acquisition”. In this case, the processor device **12** functions as a special-light image acquisition unit. The processor device **12** obtains a signal of the specific wavelength range by performing calculation based on color information of red, green, and blue or color information of cyan, magenta, and yellow included in the normal-light image.

(213) Note that red, green, and blue are sometimes referred to as RGB. In addition, cyan, magenta, and yellow are sometimes referred to as CMY.

(214) Generation Example of Feature-Quantity Image

(215) As the medical image, a feature-quantity image may be generated by using an operation based on at least any of a normal-light image obtained by radiating light in the white range or light in a plurality of wavelength ranges as the light in the white range or a special-light image obtained by radiating light in the specific wavelength range.

(216) Application Example to Program for Causing Computer to Function as Image Processing Apparatus

(217) The above-described image processing method can be configured as a program that implements functions corresponding to respective steps of the image processing method using a computer. For example, a program that causes a computer to implement an endoscopic image acquisition function, an image processing function, a display signal transmission function, and a storage function may be configured. The image processing functions may include a classification function and a region image generation function.

(218) A program that causes a computer to implement the above-described image processing function may be stored on a computer-readable information storage medium which is a non-transitory tangible information storage medium, and the program may be provided using the

information storage medium.

(219) In addition, instead of the configuration in which the program is stored on a non-transitory information storage medium and is provided, a configuration in which a program signal is provided via a network may be employed.

Combination of Embodiments, Modifications, Etc

(220) The constituent elements described in the embodiments above and the constituent elements described in the modifications can be appropriately used in combination, and some of the constituent elements can be replaced.

(221) Application Examples to Other Devices

(222) In the embodiments described above, an endoscopic image is used as an example of a medical image. However, automatic classification described in the present embodiments is also applicable to a medical image acquired using a CT apparatus, an MRI apparatus, an X-ray imaging apparatus, or the like. The medical image described in the embodiments is an example of a captured image.

(223) In the embodiments of the present invention described above, the constituent elements can be appropriately changed, added, or deleted within a scope not departing from the gist of the present invention. The present invention is not limited to the embodiments described above, and various modifications can be made by a person having the ordinary skill in the art within the technical spirit of the present invention.

REFERENCE SIGNS LIST

(224) **9** endoscope system **10** endoscope **11** light source device **12** processor device **13** display device **14** medical image processing apparatus **14a** medical image processing apparatus **15** input device **16** monitor device **20** insertion section **21** operation section **22** universal cord **25** soft part **26** bending part **27** tip part **27a** tip surface **28** imaging element **29** bending operation knob **30** air/water supply button **31** suction button **32** still image capturing instruction part **33** treatment tool introduction port **35** light guide **36** signal cable **37a** connector **37b** connector **38** endoscopic image **38a** moving image **38b** frame image **39** still image **40** image acquisition unit **41** storage instruction acquisition unit **42** image processing unit **44** display control unit **46** storage unit **46a** endoscopic image storage unit **46b** classification storage unit **46c** region image storage unit **48** classification unit **50** region image generation unit **120** control unit **122** memory **124** storage device **126** network controller **128** power supply device **130** display controller **132** input/output interface **134** input controller **136** bus **140** network **200** display screen **200a** display screen **200b** display screen **200c** display screen **200d** display screen **200e** display screen **201** frame **202** endoscopic image display area **204** region image display area **206** classification result display area **206a** classification result display area **207** frame **208** region image **208a** region image **208b** region image **208c** region image **208d** region image **208e** region image **208f** region image **208g** region image **208h** region image **209** classification contribution region **209a** first classification contribution region **209b** second classification contribution region **210** classification contribution corresponding region **210a** region to be set as classification contribution corresponding region **210b** region to be set as classification contribution corresponding region **210c** classification contribution corresponding region **210d** classification contribution corresponding region **210e** classification contribution corresponding region **210f** classification contribution corresponding region **210g** first classification contribution corresponding region **210h** second classification contribution corresponding region **211a** central portion **211b** peripheral portion **211c** intermediate region **212** text information **212a** text information **212b** text information **212c** text information **212d** text information **212e** first text information **212f** second text information **213a** first lesion **213b** second lesion **220** comparative screen **222** tumor **230** region **300** classification unit **302** CNN **304** error calculation unit **306** parameter updating unit **310** input layer **312** intermediate layers **314** output layer **320** convolutional layer **322** pooling layer **324** fully connected layer **330** information **330a** information of input layer **330b** information of intermediate layer **330c** information of fully connected layer **370** correct answer data **380** image to be processed **381a** one animal **381b** another (other) animal **382** first

classified image **384** region serving as basis of classification **386** second classified image **388**
region serving as basis of classification **390** image to be classified **391a** person **391b** animal **392**
classified image **394** person region **396** animal region **400** display screen **400a** display screen **402**
endoscopic image display area **404** region image display area **406** classification result display area
408 region image **410** classification contribution corresponding region **412** text information **412a**
text information **S10** to **S22** steps of medical image processing method

Claims

1. A medical image processing apparatus comprising one or more processors configured to: acquire an endoscopic image generated through imaging of a living body; perform classification of lesion regions contained in the endoscopic image into two or more classes; identify a classification contributing region in the endoscopic image, the classification contributing region contributing, with a degree of contribution, to the classification of one of the lesion regions contained in the endoscopic image; generate a region image displaying a region in the endoscopic image, the region image displaying the classification contributing region with a density or a heat map according to the degree of contribution; and display the region image along with the endoscopic image on a monitor, the region image being displayed at a position different from a position at which the endoscopic image is displayed.
2. The medical image processing apparatus according to claim 1, wherein the one or more processors are further configured to cause the monitor to display text information of a result of the classification.
3. The medical image processing apparatus according to claim 2, wherein the one or more processors are further configured to: in response to an area of the classification contributing region being less than a predetermined threshold, not display the text information of the result of the classification.
4. The medical image processing apparatus according to claim 1, wherein the one or more processors are further configured to perform the classification of the lesion regions contained in the endoscopic image into a neoplastic lesion and a non-neoplastic lesion.
5. The medical image processing apparatus according to claim 1, wherein the one or more processors are further configured to: perform the classification of the endoscopic image on the basis of a feature quantity acquired from the endoscopic image; and generate the region image on the basis of the feature quantity.
6. The medical image processing apparatus according to claim 1, wherein the one or more processors are further configured to: calculate, on the basis of the region image, a degree of reliability of a result of the classification; and cause the monitor to display the degree of reliability.
7. The medical image processing apparatus according to claim 1, wherein the endoscopic image is a normal-light image acquired by radiating light in a white range or light in a plurality of wavelength ranges as the light in the white range.
8. The medical image processing apparatus according to claim 1, wherein the endoscopic image is a special-light image acquired by radiating light in a specific wavelength range.
9. The medical image processing apparatus according to claim 1, wherein the one or more processors are further configured to classify lesion types of the lesion regions contained in the endoscopic image.
10. The medical image processing apparatus according to claim 1, wherein the one or more processors are further configured to classify the lesion regions contained in the endoscopic image according to an endoscopic classification.
11. The medical image processing apparatus according to claim 1, wherein: the endoscopic image is a special-light image acquired by radiating light in a specific wavelength range; and the one or more processors are further configured to: cause the monitor to display text information of a result

of the classification; and classify lesion types of the lesion regions contained in the endoscopic image.

12. The medical image processing apparatus according to claim 1, wherein the one or more processors are further configured to: employ a deep learning device that has been trained; and generate the region image on the basis of information of an intermediate layer of the deep learning device.

13. An endoscopy system comprising: an endoscope; a light source configured to supply illumination light to the endoscope; and the medical image processing apparatus according to claim 1, wherein the one or more processors acquire the endoscopic image from the endoscope.

14. The medical image processing apparatus according to claim 1, wherein the one or more processors are further configured to: in response to an area of the classification contributing region being less than a predetermined threshold, not display the region image.

15. The medical image processing apparatus according to claim 1, wherein the one or more processors are further configured to: in response to a plurality of classification contribution regions being identified and in response to an area of each of the plurality of classification contribution regions being greater than or equal to a predetermined threshold, not display the region image.

16. A medical image processing method comprising: acquiring an endoscopic image generated through imaging of a living body; performing classification of lesion regions contained in the endoscopic image into two or more classes; identifying a classification contributing region in the endoscopic image, the classification contributing region contributing, with a degree of contribution, to the classification of one of the lesion regions contained in the endoscopic image; generating a region image displaying a region in the endoscopic image, the region image displaying the classification contributing region in a density or a heat map according to the degree of contribution; and displaying the region image along with the endoscopic image on a monitor, the region image being displayed at a position different from a position at which the endoscopic image is displayed.

17. The medical image processing method according to claim 16, further comprising causing the monitor to display text information of a result of the classification.

18. The medical image processing method according to claim 16, further comprising performing the classification of the lesion regions contained in the endoscopic image into a neoplastic lesion and a non-neoplastic lesion.

19. The medical image processing method according to claim 16, further comprising: performing the classification of the endoscopic image on the basis of a feature quantity acquired from the endoscopic image; and generating the region image on the basis of the feature quantity.

20. The medical image processing method according to claim 16, further comprising: calculating, on the basis of the region image, a degree of reliability of a result of the classification; and causing the monitor to display the degree of reliability.

21. The medical image processing method according to claim 16, wherein the endoscopic image is a normal-light image acquired by radiating light in a white range or light in a plurality of wavelength ranges as the light in the white range.

22. The medical image processing method according to claim 16, wherein the endoscopic image is a special-light image acquired by radiating light in a specific wavelength range.

23. The medical image processing method according to claim 16, further comprising classifying lesion types of the lesion regions contained in the endoscopic image.

24. The medical image processing method according to claim 16, further comprising classifying the lesion regions contained in the endoscopic image according to an endoscopic classification.

25. The medical image processing method according to claim 16, wherein: the endoscopic image is a special-light image acquired by radiating light in a specific wavelength range; and the method further comprises: causing the monitor to display text information of a result of the classification; and classifying lesion types of the lesion regions contained in the endoscopic image.

26. The medical image processing method according to claim 16, further comprising: employing a

deep learning device that has been trained; and generating the region image on the basis of information of an intermediate layer of the deep learning device.
